

Regression Analysis: Group Communication Apprehension and Public Transportation Usage

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Research Question

Does group communication apprehension correlate with public transportation usage?

Answer, Response, & Summary of Results

Yes, there is a correlational relationship between group communication apprehension and public transportation usage based on the merged survey data from Prolific demographic files and Qualtrics survey responses.

Merging Datasets

```
# Combine Prolific datasets
prolific_combined = pd.concat([prolific_a, prolific_b], ignore_index=True)

# Rename participant ID column for merging
prolific_combined = prolific_combined.rename(columns={'Participant id': 'participant_id'})
qualtrics = qualtrics.rename(columns={'Q0': 'participant_id'})

# Merge datasets on participant ID
merged_data = pd.merge(qualtrics, prolific_combined, on='participant_id', how='inner')

print(f"Merged dataset shape: {merged_data.shape}")
print(f"Number of participants: {merged_data['participant_id'].nunique()}")
```

Merged dataset shape: (252, 37)

Number of participants: 252

Variables

```
# Likert scale mapping (for communication apprehension)
likert_mapping = {
    'Strongly disagree': 1,
    'Somewhat disagree': 2,
    'Neither agree nor disagree': 3,
    'Somewhat agree': 4,
    'Strongly agree': 5
}

# transportation usage
usage_mapping = {
    'Never': 0,
    '0-1 days a month': 1,
```

```

    '2-4 days a month': 2,
    '4-8 days a month': 3,
    '8 or more days a month': 4
}

# Convert communication apprehension items (Q1-Q6) to numeric
ca_items = ['Q1', 'Q2', 'Q3', 'Q4', 'Q5', 'Q6']
for item in ca_items:
    if item in merged_data.columns:
        merged_data[item + '_num'] = merged_data[item].map(likert_mapping)

# Calculate communication apprehension mean score
ca_numeric_cols = [item + '_num' for item in ca_items if item + '_num' in merged_data.columns]
merged_data['ca_mean'] = merged_data[ca_numeric_cols].mean(axis=1, skipna=True)

# Convert transportation usage items (Q26-Q29) to numeric
pt_items = ['Q26', 'Q27', 'Q28', 'Q29']
for item in pt_items:
    if item in merged_data.columns:
        merged_data[item + '_num'] = merged_data[item].map(usage_mapping)

# Calculate public transportation usage mean score
pt_numeric_cols = [item + '_num' for item in pt_items if item + '_num' in merged_data.columns]
merged_data['pt_usage_mean'] = merged_data[pt_numeric_cols].mean(axis=1, skipna=True)

print(f"Communication apprehension mean: {merged_data['ca_mean'].mean():.2f}")
print(f"Public transportation usage mean: {merged_data['pt_usage_mean'].mean():.2f}")

```

Communication apprehension mean: 3.18
Public transportation usage mean: 1.82

Regression Analysis

Primary Question: Communication Apprehension predicting Public Transportation Usage

Remaining Questions

- **Causality:** This analysis establishes correlation but cannot determine causality. Nature vs. Nurture: Does higher communication apprehension actually lead to reduced public transportation use, or do individuals with different transportation patterns develop different communication styles?
- **Measurement Validity:** The communication apprehension scale needs validation to ensure it reliably measures interest or comfortability in group settings.
- **Confounding Variables:** Additional control variables (urban/rural, income, accessibility needs, prevalence of transportation network) may influence individual's communication patterns and transportation choices.
- **Generalizability:** The sample may have selection bias from Prolific participants, which limits generalizability.

```

regression_data = merged_data[['ca_mean', 'pt_usage_mean']].dropna()

# Run regression model
if len(regression_data) > 0 and 'ca_mean' in regression_data.columns and 'pt_usage_mean' in regression_data:
    X = regression_data['ca_mean'].values
    y = regression_data['pt_usage_mean'].values

    # linear regression
    slope, intercept, r_value, p_value, std_err = stats.linregress(X, y)
    r_squared = r_value**2

    # results
    print("### Regression Results")
    print(f"R-squared: {r_squared:.3f}")
    print(f"Sample size: {len(y)}")
    print(f"Slope: {slope:.4f}")
    print(f"Intercept: {intercept:.4f}")
    print(f"p-value: {p_value:.4f}")
    print(f"Standard error: {std_err:.4f}")
else:
    print("*Regression analysis requires actual data files to run properly.*")

### Regression Results
R-squared: 0.063
Sample size: 252
Slope: 0.9562
Intercept: -1.2223
p-value: 0.0001
Standard error: 0.2339

```

Figure 1